

Intergenerational Housing Misallocation and Fertility

Compact Analytical Model

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Abstract

Children take up space, and the space they take up is scarce in a particular way: family-sized units are concentrated in the owner-occupied segment of the housing market, while buying owner housing requires a down payment and debt-service capacity that many young households do not have. This note develops a compact stationary two-period OLG theory of that friction. Each period, young households decide whether to enter a housing market, choose between renting and owning, choose how many children to have, and save for old age; births, together with a renewal inflow, generate the next young cohort. Young owners become next period's old incumbents, who decide how much of their carried housing to keep before exiting. The model delivers three results. First, every constraint on family-sized space—rental size menus, down payments, payment-to-income limits—collapses into a single object, an effective family-housing cap, and a binding cap raises the price of a child through the space the child requires. Second, the stationary competitive equilibrium clears the housing market and still misallocates the existing stock whenever young households are priced out of owner housing by financing constraints while old incumbents are locked in place by capital-gains taxes that sale triggers and death forgives; both gaps are measurable, one from household bundles and one from deeds and tax schedules. Third, at fixed prices, a housing policy changes aggregate fertility through constrained exposure, pass-through to the binding cap, the household fertility response, entry, and tenure switching.

The market structure follows [Coven et al. \(2025\)](#): tenure shapes which size menus and financing constraints a household faces, while all floorspace clears in a single market; supply margins are deferred to an extension. Children require housing space, so anything that rations family-sized space raises the private price of children. Family-sized space is rationed by tenure: by the size composition of the rental stock on one side, and by down-payment and payment-to-income requirements on the other. The old-side force—capital-gains taxes that fall due on sale but are forgiven at death—amplifies the scarcity but is not needed for any of the young-side results.

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Part I

Compact Analytical Model

1 Environment

Agents. The economy is populated by overlapping generations of young and old households. A young household has type

$$i = (y_i, a_i, B_i),$$

where $y_i > 0$ is disposable nonhousing income, $a_i \geq 0$ is liquid wealth, and $B_i \geq 0$ is the gross bequest assigned at entry, taxed at rate τ^b on receipt. It decides whether to enter the housing market and, inside, chooses tenure, occupied housing, and fertility. Old households come in two groups. Old incumbents are last period's young owners, with states

$$j = (a_j, H_j^0, \ell_j, \mathcal{B}_j),$$

where a_j is financial wealth at the start of old age (negative values are outstanding mortgage debt), $H_j^0 > 0$ is the housing carried into old age, and $\ell_j \geq 0$ is the capital-gains levy per unit of that housing sold while alive; they choose consumption, how much housing to retain, and how large an estate to leave, then exit. Old renters are last period's young renter entrants, with states $j = (a_j, \mathcal{B}_j)$, $a_j > 0$; they rent in old age, carry no housing and no levy, and exit. Nonentrants live outside the modeled market in both periods.

Masses. Time is discrete. At the start of period t a mass $M_t > 0$ of potential young households is drawn from a distribution $G_{Y,t}$ over types. Next period's potential young are the maturing children of the current cohort plus a renewal inflow of mass R_t ; entry then determines how many of them participate in the housing market. The period- t measures of old incumbents and old renters are $F_{O,t}$ and $F_{R,t}$. Let $\bar{M}$ denote the stationary cohort mass. In a stationary equilibrium, $G_{Y,t} = G_Y$, $F_{O,t} = F_O$, and $F_{R,t} = F_R$ are reproduced by the transition laws below.

Preferences. Households have preferences over consumption, housing, and children. Children raise the household's consumption and housing needs: each child claims $\chi > 0$ units of the consumption good and $\kappa > 0$ units of housing services, so with consumption spending c_i , occupied housing h_i , and fertility n_i , young preferences are

$$u_i^Y(c_i, h_i, n_i) = \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \beta \log n_i, \quad \alpha, \beta > 0.$$

One more child means less space and less consumption left for the parents: children compete with their parents for the same house and the same budget. Rearranging spending on consumption and housing at user cost q shows the price of that competition,

$$(c_i - \chi n_i) + q(h_i - \kappa n_i) + (\chi + \kappa q)n_i = c_i + qh_i :$$

the full price of a child is its consumption requirement plus the market value of the space it occupies. When access to space is constrained, the space component carries a premium. A useful benchmark is equal scaling, $\chi = \kappa q / \alpha$, under which one child uses consumption and housing in the same proportions as its parents. Because $\beta \log n_i$ rules out $n_i = 0$, every entered household has positive completed fertility: the model studies the intensive margin, not childlessness.

Old preferences are

$$u_j^O(c_j^O, h_j^O, e_j) = \log c_j^O + \gamma \log h_j^O + \mathcal{B}_j(e_j), \quad \gamma > 0,$$

over consumption, retained housing, and the estate $e_j \geq 0$ the household leaves when it exits, valued through a warm-glow function $\mathcal{B}_j$, increasing and concave, possibly identically zero, and itself part of the old type: bequest motives are heterogeneous. The exiting cohort's estates fund the bequests $B_{i'}$ received by the cohort that arrives, through the transition below. A young household also values its own old age: its objective adds $\vartheta \mathbb{E} u_{j'}^O$, evaluated at the old-age choices it will make, with discount factor $\vartheta \in (0, 1]$ and the expectation over the old type assigned by the transition below.

Housing. There is a fixed aggregate stock $\bar{H}$ of divisible floorspace, measured in units of housing services. A household occupies floorspace in one of two tenure modes: renting, denoted R , or owning, denoted O . The modes offer different size menus,

$$\mathcal{H}^R = \{h : 0 < h \leq \bar{h}^R\}, \quad \mathcal{H}^O = \{h : 0 < h \leq \bar{h}^O\}, \quad 0 < \bar{h}^R < \bar{h}^O.$$

Family-sized houses are mostly available to buy, not to rent: a household that wants substantially more space than $\bar{h}^R$ must buy it.¹ All occupied floorspace trades in one market. The asset price of a unit of floorspace is P , and a no-arbitrage user-cost condition ties the rental rate to the price:

$$q = \rho P, \quad \rho = r + \delta + \tau^p, \quad (1)$$

where r is the interest rate, δ is depreciation, and τ^p is a recurring property tax. Renters and owner-occupiers therefore face the same per-unit flow cost q on new occupancy choices; tenure matters through the size menus and through financing, not through the per-unit price of space, and the capital-gains levy below shifts only old incumbents' retention margin. Given q , a higher property tax lowers the asset price P and hence the down payment needed to buy a given amount of space. At the start of a stationary period, old incumbents carry into old age the houses generated by previous young owner choices, and passive landlords hold the residual stock:

$$H_{\text{pass}} = \bar{H} - \int H_j^0 dF_O(j) \geq 0.$$

In a stationary equilibrium the inequality is implied by market clearing: the previous cohort's owner purchases cannot exceed the stock. Housing payments are transfers across households, landlords, and fiscal accounts, closed by lump-sum rebates, and reallocating the existing stock costs no real resources.²

Financing. Buying floorspace requires financing. A young buyer can finance at most a share $\phi \in (0, 1)$ of the purchase and faces a payment-to-income limit $\psi \in (0, 1)$ on housing outlays. Pledgeable resources for the down payment are

$$A_i(x) = a_i + xB_i + T_i^b(x), \quad x = 1 - \tau^b,$$

where τ^b is an estate tax and $T_i^b(x)$ is the estate-tax rebate assigned to type i . The down payment is posted at purchase, at the start of the period, out of wealth in hand: income accrues over the period,

¹Coven et al. (2025) model the same segmentation as a minimum size for owned homes, with rentals available in any size; we cap rental size instead. Both make family-sized space owner-concentrated; ours keeps the renter problem interior at the bottom.

²Two supply extensions are deliberately left out of the baseline and developed later: a price-elastic aggregate supply curve as in Coven et al. (2025), and tenure-specific stocks with separate construction margins, under which the rental and owner user costs diverge endogenously. The local results below survive with q replaced by the relevant tenure-specific user cost.

so $A_i(x)$ caps what a buyer can put down. Wealth is otherwise fungible. It enters the young budget below alongside income and finances consumption and saving; buying assets is not expenditure, so the posted equity remains part of the household's wealth. Inheritances and the estate tax therefore move both access to owner housing and lifetime resources. Households save: a young household carries $a'_i \geq 0$ into old age at the exogenous gross return $1 + r$, and cannot borrow unsecured against old age. Newborn households inherit the estates of the old who exit as they arrive: B_i is assigned from old estates by the young-type transition below, and the estate tax falls on these transfers. The interest rate r is exogenous: households borrow and lend with a passive financial sector, so no domestic capital-market clearing condition is imposed, and the only price determined inside the model is q .

Capital-gains taxation. A housing sale during life realizes the seller's accrued capital gain, taxed at a proportional rate above an exclusion. Housing held until death passes to heirs with a stepped-up basis, and the accrued gain is never taxed. For an old incumbent the difference between the two disposal routes is a levy $\ell_j \geq 0$ per unit of housing sold while alive: the gains-tax rate times her accrued per-unit gain when that gain exceeds the exclusion, and zero otherwise. With a constant aggregate price, accrued gains reflect idiosyncratic repricing histories, and the old-incumbent transition carries their stationary cross-section; at the purchase margin the young buyer treats this repricing as neutral in expectation (Section 7).³

Demography and transitions. The period timing is: potential young types are drawn; young households decide whether to enter; entrants choose tenure, housing, fertility, and savings; old incumbents choose consumption, retained housing, and estates; old households exit; births and the renewal inflow form the next potential-young cohort; current young owners become next period's old incumbents. Let

$$\Gamma_O(dj' \mid i, h, a'; q, \tau^p)$$

be the old-incumbent transition for a current young owner of type i who buys h and carries savings a' . It generates

$$j' = (a_{j'}, H_{j'}^0, \ell_{j'}, \mathcal{B}_{j'}).$$

The housing component is $H_{j'}^0 = h$, and the wealth component settles the household's books at the market return:

$$a_{j'} + qH_{j'}^0 = (1 + r)a' + g_{j'}h,$$

where $g_{j'}$ is the idiosyncratic per-unit repricing of the household's units between youth and old age, drawn by the kernel together with the levy it generates: $\ell_{j'}$ is the gains-tax rate times $g_{j'}$ above the exclusion, zero below. The left side is total old wealth, financial plus housing, both in per-period goods units: the old live one more period, so a carried unit is worth one period of services q to its holder, and the shell that outlives her recirculates through the stock. The right side is what the household's savings earn plus what its units gained. The asset price P appears only in the down-payment requirement. Financing positions from youth settle inside $a_{j'}$: a household that carried a mortgage has lower financial wealth and the same total. The draw does not condition on fertility: estates go to the arriving cohort, not to own children, so the model has no old-age-support motive for births. Young renter entrants map into old-renter states through a parallel kernel $\Gamma_R(dj' \mid i, a')$, which assigns $\mathcal{B}_{j'}$ and $a_{j'} = (1 + r)a'$; nonentrants receive zero mass under both kernels. Old-renter states follow the same law as (2) with Γ_R and the renter indicator. Given current

³Assessment protection of the Proposition 13 type, under which incumbents are taxed on below-market assessed values until they move, generates the same per-unit retention wedge.

young choices, the next old-incumbent measure satisfies

$$F_{O,t+1}(A) = M_t \int \pi_{i,t}^E \mathbf{1}\{o_{i,t} = O\} \Gamma_O(A \mid i, h_{i,t}^O, a_{i,t}^{'O}; q_t, \tau_t^p) dG_{Y,t}(i) \quad (2)$$

for every measurable set A of old states. In stationarity, this measure equals F_O .

Let

$$n_{i,t}^Y = \mathbf{1}\{o_{i,t} = R\} n_{i,t}^R + \mathbf{1}\{o_{i,t} = O\} n_{i,t}^O \quad (3)$$

denote fertility in the chosen inside branch, and let $\bar{n}_i^E \geq 0$ denote fertility in the outside option. The baseline formulas set $\bar{n}_i^E = 0$, but the demographic law allows outside fertility or inflow to renew the population.

Young type $i' = (y_{i'}, a_{i'}, B_{i'})$ is generated by fertility, old bequests, and the renewal inflow. Let

$$\Gamma_Y(d i' \mid i, m, n, F_O, F_R, e; \Upsilon), \quad m \in \{R, O, E\},$$

be the young-type kernel for a birth associated with current type i , branch m , fertility n , old bequest policy e , and renewal law Υ . The old who exit at the end of period t endow the cohort that arrives at $t + 1$: the kernel assigns their bequest expenditure as the entrants' received bequests $B_{i'}$ and supplies the noninheritance components of young types. In stationarity, Γ_Y is restricted so that assigned bequests aggregate to the estates of all exiting old households, incumbents and renters:

$$\bar{M} \int \left[\pi_i^E n_i^Y \mathbb{E}[B_{i'} \mid i, o_i] + (1 - \pi_i^E) \bar{n}_i^E \mathbb{E}[B_{i'} \mid i, E] \right] dG_Y(i) = \int e_j d(F_O + F_R)(j),$$

where the expectations are over Γ_Y and renewal draws from Υ receive no bequests. Let $R_t \geq 0$ be the renewal inflow, outside births or immigration, with type law Υ . The next young cohort satisfies, for every measurable set A of young types,

$$\begin{aligned} M_{t+1} G_{Y,t+1}(A) = M_t \int & \left[\pi_{i,t}^E n_{i,t}^Y \Gamma_Y(A \mid i, o_{i,t}, n_{i,t}^Y, F_{O,t}, F_{R,t}, e_t; \Upsilon) \right. \\ & \left. + (1 - \pi_{i,t}^E) \bar{n}_i^E \Gamma_Y(A \mid i, E, \bar{n}_i^E, F_{O,t}, F_{R,t}, e_t; \Upsilon) \right] dG_{Y,t}(i) + R_t \Upsilon(A). \end{aligned} \quad (4)$$

Setting A equal to the whole young type space gives the cohort-mass law:

$$M_{t+1} = M_t \int \left[\pi_{i,t}^E n_{i,t}^Y + (1 - \pi_{i,t}^E) \bar{n}_i^E \right] dG_{Y,t}(i) + R_t. \quad (5)$$

In a stationary equilibrium, $M_t = \bar{M}$, $G_{Y,t} = G_Y$, $F_{O,t} = F_O$, and $R_t = \bar{R}$. If $\bar{R} = 0$, stationarity requires average births per potential young household to be one. If $\bar{R} > 0$, it requires average births below one, with the shortfall covered by renewal.

Entry. A potential young household can enter the housing market or take an outside option with lifetime value $\bar{W}^E$; entry is subject to Type-I extreme-value taste shocks with scale $\kappa_E > 0$.

2 The young household's problem

Conditional on entry, household i takes the user cost q and the asset price $P = q/\rho$ as given and chooses a tenure mode $m \in \{R, O\}$, a bundle, and savings. The two modes share one objective

and one budget; they differ in the size menu and in whether the financing constraints apply. For $m \in \{R, O\}$,

$$W_i^m(q, \tau^p, x) = \max_{c_i, h_i, n_i, a'_i} \left\{ \log(c_i - \chi n_i) + \alpha \log(h_i - \kappa n_i) + \beta \log n_i + \vartheta \mathbb{E}[V_j^O \mid i, m, h_i, a'_i] \right\} \quad (6)$$

subject to

$$c_i + qh_i + a'_i = y_i + A_i(x), \quad (7)$$

$$h_i \in \mathcal{H}^m, \quad a'_i \geq 0,$$

$$\mathbf{1}\{m = O\} (1 - \phi) Ph_i \leq A_i(x), \quad (8)$$

$$\mathbf{1}\{m = O\} qh_i \leq \psi y_i, \quad (9)$$

$$c_i - \chi n_i > 0, \quad h_i - \kappa n_i > 0.$$

V_j^O is the old-age value defined in the next section; the expectation is over the old type drawn by Γ_O for owners and by Γ_R for renters, given the household's choices. For $m = R$ the indicator constraints hold vacuously: a renter faces only the rental menu. The budget is the same in both modes—space costs q per unit either way, wealth is spendable either way, savings earn $1 + r$ either way—so tenure matters through the menu and the two owner financing requirements, not through prices.

Using $P = q/\rho$, owner housing is bounded above by

$$h_i \leq H_i^O(q, \tau^p, x) = \min \left\{ \bar{h}^O, \frac{A_i(x)\rho}{(1 - \phi)q}, \frac{\psi y_i}{q} \right\}.$$

The renter analog is $H_i^R = \bar{h}^R$. In mode m , H_i^m is the effective family-housing cap. A renter is capped by the rental menu. A would-be buyer is capped by whichever of the owner menu, down-payment capacity, or payment-to-income capacity is shortest. A policy affects fertility through this channel only when it relaxes the binding component of H_i^m . If the owner branch has no feasible allocation, for example when $A_i(x) = 0$, set $W_i^O = -\infty$.

The inside value of entry is

$$W_i^Y(q, \tau^p, x) = \max\{W_i^R(q), W_i^O(q, \tau^p, x)\}. \quad (10)$$

Let $o_i \in \{R, O\}$ denote an optimal tenure.⁴

3 The old household's problem

The baseline young-access mechanism below does not require any old-side force. When the old side is active, the compact model uses the capital-gains institution from the environment. On old-household objects (V_j^O , u_j^O , c_j^O , MV_j^O) the superscript marks the old stage, not owner status; the index j separates them from young owner objects. Old incumbent j is last period's young owner and enters with financial wealth a_j , carried housing H_j^0 , and an accrued gain on it. Each unit she holds faces one choice: sell it now and pay the gains tax, or keep it until death, when the basis

⁴The hard rental cap $\bar{h}^R$ is the simplest representation of size segmentation. A smooth alternative lets renters pay $R(h) = qh + \varrho(h)$ with $\varrho_h(h) \geq 0$ for large units, so that large rentals exist but at a premium; every marginal condition below goes through with q replaced by $R_h(h)$. Such a premium is also one natural source of a rent-own gap in per-unit costs, which Section 9 studies.

steps up and the tax is forgiven. The levy ℓ_j is the tax per unit that selling triggers, zero when her gain sits below the exclusion.⁵ She chooses consumption, retained housing, and her estate, with the preferences of Section 1:

$$V_j^O = \max_{c_j^O, h_j^O, e_j} u_j^O(c_j^O, h_j^O, e_j) \text{ subject to } c_j^O + e_j + q h_j^O = a_j + q H_j^0 - \ell_j(H_j^0 - h_j^O), \quad 0 \leq h_j^O \leq H_j^0. \quad (11)$$

Releasing a unit of housing nets $q - \ell_j$, its market value less the tax the sale realizes, while the market values the released space at q . The forgiveness at death is a subsidy to staying put: a tax rule attached to the disposal route, not a preference and not a real resource saving. The planner below counts true old housing utility and true warm-glow bequest utility but does not count the avoided tax as a social benefit. An old household that keeps a large house because she values it is using housing efficiently; an old household that keeps it because selling would realize a taxable gain is not.

An old renter carries no housing and no levy: her wealth is her returned savings, $a_j = (1 + r)a'_i$, positive because zero savings would leave her nothing to consume. With the same preferences she solves

$$V_j^O = \max_{c_j^O, h_j^O, e_j} u_j^O(c_j^O, h_j^O, e_j) \text{ subject to } c_j^O + e_j + q h_j^O = a_j, \quad 0 < h_j^O \leq \bar{h}^R, \quad (12)$$

and at an interior choice $\gamma c_j^O / h_j^O = q$: old rental demand adds to market clearing but carries no wedge.⁶

4 Entry

With the entry technology of Section 1, the probability that type i enters the housing market is

$$\pi_i^E(q, \tau^p, x) = \frac{\exp\{W_i^Y(q, \tau^p, x)/\kappa_E\}}{\exp\{W_i^Y(q, \tau^p, x)/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}. \quad (13)$$

The endogenous measure of young entrants is

$$dF_Y^E(i; q, \tau^p, x) = \bar{M} \pi_i^E(q, \tau^p, x) dG_Y(i).$$

The limit $\kappa_E \downarrow 0$ gives deterministic entry, with the indifference set $W_i^Y = \bar{W}^E$ assumed to carry zero mass. Conditional on entry, the household chooses tenure according to (10).

5 Competitive equilibrium

A stationary equilibrium clears one aggregate floorspace market and closes the old-incumbent, young-type, and cohort-mass fixed points. Households take the user cost as given, choose tenure and housing, and their choices aggregate into demand for the fixed stock. There is no pairwise

⁵The statutory tax is triggered by the sale of the discrete dwelling; with divisible floorspace the per-unit levy is its smooth counterpart. Discreteness concentrates the same wedge at the downsizing margin and strengthens retention.

⁶Coven et al. (2025) model this lock-in channel in lifecycle form and evaluate a rate reduction. The levy ℓ_j prices the same margin statically, and also prices the opposite reform: taxing accrued gains at every transfer, including bequest, removes the step-up and shrinks ℓ_j to the value of tax deferral.

assignment of old units to young households: one price equates total demand to the stock, and nothing in that condition asks whether the marginal family-sized unit is held by the household that values it most. That question is deferred to the efficiency analysis below.

Given policies (τ^p, x) , the stock $\bar{H}$, the stationary cohort mass $\bar{M}$, young distribution G_Y , old measures F_O and F_R , and transfer rules, aggregate floorspace demand $H^D(q, \tau^p, x; \bar{M}, G_Y, F_O, F_R)$ at a candidate user cost q adds young households, old incumbents, and old renters:

$$H^D = \bar{M} \int \pi_i^E [\mathbf{1}\{o_i = R\} h_i^R + \mathbf{1}\{o_i = O\} h_i^O] dG_Y(i) + \int h_j^O dF_O(j) + \int h_j^O dF_R(j).$$

Tenure determines which menu and which financing constraints apply to a household, not which market it buys space in: all of this demand falls on the same stock.

Definition 1 (Stationary competitive equilibrium). *Given $(\tau^p, x, \bar{H})$, transition laws $(\Gamma_O, \Gamma_R, \Gamma_Y)$, and a stationary renewal law $(\bar{R}, \Upsilon)$, a stationary competitive equilibrium consists of a stationary cohort mass $\bar{M} > 0$, a user cost q , an asset price P , a stationary young distribution G_Y , a stationary old-incumbent measure F_O , a stationary old-renter measure F_R , young branch allocations including savings, tenure choices, entry probabilities, old allocations, and fiscal transfers such that:*

1. *conditional on entry, young households solve (6) in each mode, then choose tenure by (10);*
2. *old incumbents solve (11) and old renters solve (12);*
3. *entry is given by (13);*
4. *the asset price satisfies the no-arbitrage condition (1);*
5. *the single floorspace market clears,*

$$H^D(q, \tau^p, x; \bar{M}, G_Y, F_O, F_R) = \bar{H}; \quad (14)$$

6. *passive housing-income, credit, property-tax, capital-gains, estate-tax, and policy-instrument accounts are closed by lump-sum transfers;*
7. *the old-incumbent measure is stationary: for every measurable set A of old states,*

$$F_O(A) = \bar{M} \int \pi_i^E \mathbf{1}\{o_i = O\} \Gamma_O(A \mid i, h_i^O, a_i'^O; q, \tau^p) dG_Y(i), \quad (15)$$

and the old-renter measure satisfies the same condition with Γ_R and $\mathbf{1}\{o_i = R\}$;

8. *the young cohort is stationary: for every measurable set A of young types,*

$$\begin{aligned} \bar{M} G_Y(A) = \bar{M} \int & \left[\pi_i^E n_i^Y \Gamma_Y(A \mid i, o_i, n_i^Y, F_O, F_R, e; \Upsilon) \right. \\ & \left. + (1 - \pi_i^E) \bar{n}_i^E \Gamma_Y(A \mid i, E, \bar{n}_i^E, F_O, F_R, e; \Upsilon) \right] dG_Y(i) + \bar{R} \Upsilon(A), \end{aligned} \quad (16)$$

where e is the old bequest-expenditure policy. Taking A to be the whole young type space gives the stationary cohort-balance condition

$$\bar{M} = \bar{M} \int [\pi_i^E n_i^Y + (1 - \pi_i^E) \bar{n}_i^E] dG_Y(i) + \bar{R}. \quad (17)$$

Let $\Xi^\Pi \geq 0$ denote real resource costs of policy instruments. The transfer rule is chosen so that, after summing household budgets and using (14), the aggregate goods constraint is

$$\begin{aligned} \bar{M} \int \pi_i^E [\mathbf{1}\{o_i = R\}(c_i^R + a_i'^R - y_i - A_i(x)) + \mathbf{1}\{o_i = O\}(c_i^O + a_i'^O - y_i - A_i(x))] dG_Y(i) \\ + \int (c_j^O + e_j - a_j) d(F_O + F_R)(j) + \Xi^\Pi \leq 0. \end{aligned}$$

Housing quantities are disciplined separately by floorspace feasibility: housing payments are transfers that net out against rebates, and the old carried-housing term qH_j^0 is a claim on the existing stock rather than new output. The intergenerational books close through the transitions: bequest receipts net against old estates by the adding-up restriction on Γ_Y , estate-tax collections net against the rebates T_i^b , and the net return r on carried savings is paid by the passive financial sector, the open-economy counterpart of foreign asset income. Housing in this baseline is a pure allocation problem.

Aggregate births generated by the potential-young cohort are

$$N = \bar{M} \int [\pi_i^E n_i^Y + (1 - \pi_i^E) \bar{n}_i^E] dG_Y(i).$$

In a stationary demographic equilibrium, $N + \bar{R} = \bar{M}$. Under the baseline outside-option normalization $\bar{n}_i^E = 0$, $N = \bar{M} \int \pi_i^E n_i^Y dG_Y(i)$.

6 Constrained planner

The planner comparison is evaluated at a stationary competitive equilibrium. For the local variations below, the stationary cohort mass $\bar{M}$, young distribution G_Y , old-incumbent measure F_O , transition laws $(\Gamma_O, \Gamma_R, \Gamma_Y)$, renewal inflow $\bar{R}$, and tax rules are held fixed at the instant of the variation. The question is which constraints young families face as real scarcity and which are financing arrangements that a different institutional design could undo.

The planner has the same preferences, stationary young distribution, old-incumbent measure, entry technology, housing stock, and tenure size menus as the competitive economy. It chooses entry-relevant inside allocations, tenure assignments, young consumption, housing and fertility, old consumption, housing and estates, and policy instruments for the current stationary cross-section. It does not move the savings margin: young savings a_i' , and the old wealth they generate, are held at their equilibrium values, so the comparison is within-period reallocation given the intergenerational wealth transition. The planner respects floorspace feasibility and the rental and owner menus.

The planner faces the economy's real constraints and nothing else: the floorspace stock, the tenure size menus, goods feasibility, and the domain conditions below. It does not face household budgets, since allocations are supported by lump-sum transfers, and it does not face the financing inequalities, which restrict market purchases rather than occupancy; when an allocation must be implemented through market purchases, Section 8 prices the required instruments by their marginal real cost. The planner counts true old housing utility and true warm-glow bequest utility; it does not count the avoided gains tax ℓ_j , which is a transfer.

Let $\omega_i^Y > 0$ and $\omega_j^O > 0$ be Pareto weights. Let $\mathcal{G}_i(n_i)$ denote an optional social payoff from births beyond parental utility. The fertility-neutral planner has $\mathcal{G}_i \equiv 0$.

If the planner assigns young type i tenure $o_i^P \in \{R, O\}$ and allocation (c_i, h_i, n_i) , the inside value relevant for entry is

$$V_i^P = u_i^Y(c_i, h_i, n_i).$$

The induced entry probability is

$$\pi_i^P = \frac{\exp\{V_i^P/\kappa_E\}}{\exp\{V_i^P/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}.$$

The corresponding expected entry value, up to an additive constant, is

$$\mathcal{W}_i^E(V_i^P) = \kappa_E \log [\exp\{V_i^P/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}].$$

The planner solves

$$\max \bar{M} \int \omega_i^Y \mathcal{W}_i^E(V_i^P) dG_Y(i) + \bar{M} \int \pi_i^P \mathcal{G}_i(n_i) dG_Y(i) + \int \omega_j^O u_j^O(c_j^O, h_j^O, e_j) d(F_O + F_R)(j) \quad (18)$$

subject to floorspace feasibility,

$$\bar{M} \int \pi_i^P [\mathbf{1}\{o_i^P = R\} + \mathbf{1}\{o_i^P = O\}] h_i dG_Y(i) + \int h_j^O d(F_O + F_R)(j) \leq \bar{H},$$

menu feasibility,

$$h_i \in \mathcal{H}^R \text{ if } o_i^P = R, \quad h_i \in \mathcal{H}^O \text{ if } o_i^P = O,$$

and goods feasibility,

$$\begin{aligned} & \bar{M} \int \pi_i^P (c_i + a'_i - y_i - A_i(x)) dG_Y(i) \\ & + \int (c_j^O + e_j - a_j) d(F_O + F_R)(j) + \Xi^\Pi \leq 0, \end{aligned}$$

All allocations also satisfy

$$c_i - \chi n_i > 0, \quad n_i > 0, \quad h_i - \kappa n_i > 0,$$

for young households, and

$$c_j^O > 0, \quad e_j \geq 0, \quad 0 \leq h_j^O \leq H_j^0,$$

for old incumbents, with the retention bound replaced by the rental menu for old renters.

Let Λ be the multiplier on goods feasibility and ΛQ the multiplier on floorspace feasibility, so that Q is the planner's scarcity price of space in consumption goods. For a fixed tenure assignment and active set, and away from menu corners, an interior planner allocation satisfies

$$\begin{aligned} \frac{u_{h,i}^Y}{u_{c,i}^Y} &= Q && \text{for young households,} \\ \frac{u_{h,j}^O}{u_{c,j}^O} &= Q && \text{for old housing,} \end{aligned}$$

$$\frac{\beta(c_i - \chi n_i)}{n_i} + \nu_i^g = \chi + \kappa Q,$$

$$\frac{\mathcal{B}'_j(e_j)}{u_{c,j}^O} = 1 \quad \text{if } e_j > 0.$$

The planner equalizes every household's housing-consumption tradeoff to the common scarcity price of space, prices children at $\chi + \kappa Q$, and respects the bequest motive. The term ν_i^g is the goods-equivalent social value of a marginal birth beyond parental utility. If $\mathcal{G}_i \equiv 0$, then $\nu_i^g = 0$. Endogenous logit entry affects the levels of the first-order conditions through a common factor, but that factor cancels from the housing-consumption and fertility ratios displayed here.

7 Marginal conditions in equilibrium

How tightly a constraint binds can be read off the household's own bundle. A household that is free to adjust equates its marginal value of housing space to the price of space. A household pinned by a size menu or a financing requirement values space above its price, and the distance between the two is the constraint's bite, measured in consumption goods.

For a renter, let λ_i^R be the multiplier on the budget (7) and $\theta_i^R \geq 0$ the multiplier on the rental size cap $h_i \leq \bar{h}^R$. At an interior allocation with respect to consumption and fertility,

$$\begin{aligned} \frac{1}{c_i^R - \chi n_i^R} &= \lambda_i^R, \\ \frac{\alpha}{h_i^R - \kappa n_i^R} &= \lambda_i^R q + \theta_i^R, \\ \frac{\beta}{n_i^R} &= \frac{\chi}{c_i^R - \chi n_i^R} + \frac{\alpha \kappa}{h_i^R - \kappa n_i^R}. \end{aligned} \tag{19}$$

Converting the cap multiplier into consumption goods, $\zeta_i^R = \theta_i^R / \lambda_i^R \geq 0$, the renter's marginal value of housing is

$$MV_i^R \equiv \frac{u_{h,i}^Y}{u_{c,i}^Y} = \frac{\alpha(c_i^R - \chi n_i^R)}{h_i^R - \kappa n_i^R} = q + \zeta_i^R. \tag{20}$$

A renter against the size cap values one more unit of space above the rent it would command, and the gap is computable from the observed bundle as $\zeta_i^R = \alpha(c_i^R - \chi n_i^R)/(h_i^R - \kappa n_i^R) - q$, the household's tradeoff between space and consumption, net of children, in excess of the market rent. An unconstrained renter has $\zeta_i^R = 0$.

Assumption 1 (Regular transition, repricing neutral at purchase). *Conditional on a young owner's type and savings: (i) the kernel draws of $(g_{j'}, \ell_{j'}, \mathcal{B}_{j'})$ do not vary with h_i except through the books of Section 1; (ii) the induced old retention choice is interior, $h_{j'}^O < H_{j'}^0$; (iii) the draw satisfies $\mathbb{E}[(g_{j'} - \ell_{j'})/c_{j'}^O] = 0$.*

Buying one more unit when young changes old-age wealth through the unit's repricing $g_{j'}$ and the levy $\ell_{j'}$ it generates. Parts (i) and (ii) make the envelope of the old value with respect to carried housing equal $(q - \ell_{j'})/c_{j'}^O$ —no distributional shift in the draw, no corner value on the retention bound—and part (iii) then makes the carry term vanish in expectation: anticipation of the levy does not distort the purchase margin, and the levy matters where this paper studies it, ex post, on

the realized cross-section of old incumbents. Without part (iii), the owner's housing condition below gains a carry term $\vartheta \mathbb{E}[(g_{j'} - \ell_{j'})/c_{j'}^O]$, in marginal-utility units; divided by λ_i^O it is a premium or discount on the owner's effective user cost.

For a young owner, let λ_i^O be the multiplier on the budget (7), $\mu_i \geq 0$ the multiplier on the down-payment constraint (8), $\eta_i \geq 0$ the multiplier on the payment-to-income constraint (9), and $\theta_i^O \geq 0$ the multiplier on the owner menu cap $h_i \leq \bar{h}^O$. At an interior allocation with respect to consumption and fertility,

$$\begin{aligned} \frac{1}{c_i^O - \chi n_i^O} &= \lambda_i^O, \\ \frac{\alpha}{h_i^O - \kappa n_i^O} &= \lambda_i^O q + \mu_i(1 - \phi)P + \eta_i q + \theta_i^O, \\ \frac{\beta}{n_i^O} &= \frac{\chi}{c_i^O - \chi n_i^O} + \frac{\alpha \kappa}{h_i^O - \kappa n_i^O}. \end{aligned} \quad (21)$$

The same conversion for the owner collects the three access constraints, each priced by the primitive that generates it:

$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i^O}(1 - \phi)P + \frac{\eta_i}{\lambda_i^O}q}_{\zeta_i^{O,F}, \text{ financial component}} + \underbrace{\frac{\theta_i^O}{\lambda_i^O}}_{\text{owner-menu component}} \geq 0.$$

The first term is the collateral margin: it is positive when the household's pledgeable wealth $A_i(x)$ cannot cover the down payment $(1 - \phi)P$ on a larger unit. The second is the debt-service margin: positive when the payment limit ψy_i binds. The third is the owner size menu. If the owner menu cap does not bind, then $\theta_i^O = 0$ and $\zeta_i^O = \zeta_i^{O,F}$: the household is constrained by finance, not by the stock. The young owner's marginal value of owner housing is

$$MV_i^O \equiv \frac{u_{h,i}^Y}{u_{c,i}^Y} = \frac{\alpha(c_i^O - \chi n_i^O)}{h_i^O - \kappa n_i^O} = q + \zeta_i^O. \quad (22)$$

Both tenure modes share the intertemporal condition

$$\frac{1}{c_i - \chi n_i} \geq \vartheta(1 + r) \mathbb{E} \left[\frac{1}{c_{j'}^O} \middle| i, m, h_i, a_i' \right], \quad \text{with equality if } a_i' > 0, \quad (23)$$

the Euler equation: the envelope condition on the old problem gives $\partial V_{j'}^O / \partial a_{j'} = 1/c_{j'}^O$, and carried savings pay $1 + r$ in every old state. In the benchmark the Euler equation pins savings in closed form. With logarithmic warm glow, $\mathcal{B}_j(e) = b_j \log e$ with $b_j \geq 0$, old consumption at interior choices is proportional to old wealth; if anticipated repricing is degenerate at zero, not merely neutral on average, the Euler equation collapses to the savings rule

$$a_i' = k_i(c_i - \chi n_i), \quad k_i = \vartheta(1 + \gamma + \mathbb{E}[b_{j'} \mid i]), \quad (24)$$

and the gross return drops out: with log utility the income and substitution effects of r cancel. The savings load k_i is old age's claim on each unit of adult consumption. The comparative statics of Section 9 use this rule and suppress the type index on k .

For an old owner at an interior choice of current housing to keep,

$$MV_j^O \equiv \frac{u_{h,j}^O}{u_{c,j}^O} = \frac{\gamma c_j^O}{h_j^O} = q - \ell_j. \quad (25)$$

The old owner's valuation sits *below* the market price by exactly the levy: she keeps marginal space she values at less than the market would pay, because selling it would realize the taxable gain. If bequests are interior, then

$$\mathcal{B}'_j(e_j) = \frac{1}{c_j^O},$$

which matches the planner's bequest condition: leaving wealth to children is a preference the model respects, not a distortion. An old renter at an interior choice has $MV_j^O = q$ exactly; her demand is wedge-free.

Proposition 1 (Constrained family-housing access and fertility). *Fix an entered young household and its optimal tenure branch. If the household rents, then*

$$\frac{\beta (c_i^R - \chi n_i^R)}{n_i^R} = \chi + \kappa(q + \zeta_i^R).$$

If the household owns, then

$$\frac{\beta (c_i^O - \chi n_i^O)}{n_i^O} = \chi + \kappa(q + \zeta_i^O).$$

Thus rental size limits and owner access constraints enter fertility through the housing services required per child.

Proof. For renters, multiply (19) by $1/\lambda_i^R = c_i^R - \chi n_i^R$ and use (20). For owners, multiply (21) by $1/\lambda_i^O = c_i^O - \chi n_i^O$ and use (22). $\square$

A child's space requirement is priced at the household's *own* shadow value of space, not at the market price. In an unconstrained branch, the full price of a child is $\chi + \kappa q$. For a constrained household it is $\chi + \kappa(q + \zeta_i^m)$: children become more expensive exactly where family-sized space is hard to reach, even with market prices unchanged. Two limits anchor the framework: as $H_i^m \rightarrow \infty$ every household pays the unconstrained child price $\chi + \kappa q$, and with $\kappa = 0$ children claim no space and the price reduces to the standard child-goods cost χ . On the financing side, when the collateral component binds, $\partial H_i^O / \partial \tau^p = A_i(x) / ((1 - \phi)q) > 0$: a higher property tax lowers the asset price at a given user cost and relaxes the down payment, the capitalization channel of Coven et al. (2025).

If no access or menu constraint binds in the relevant tenure branch, the corresponding access value is zero. For example, an unconstrained young owner satisfies

$$\begin{aligned} c_i^u - \chi n_i^u &= \frac{y_i + A_i(x)}{1 + \alpha + \beta + k_i}, & a_i^u &= k_i (c_i^u - \chi n_i^u), \\ n_i^u &= \frac{\beta (y_i + A_i(x))}{(1 + \alpha + \beta + k_i)(\chi + \kappa q)}, \\ h_i^u &= \frac{\alpha (y_i + A_i(x))}{(1 + \alpha + \beta + k_i)q} + \kappa \frac{\beta (y_i + A_i(x))}{(1 + \alpha + \beta + k_i)(\chi + \kappa q)}. \end{aligned}$$

The renter analog has the same closed form and is feasible only if $h_i^u \leq \bar{h}^R$.

8 Efficiency

At a stationary competitive equilibrium, the price makes aggregate demand equal the fixed stock. Market clearing alone is not enough for efficiency. Conditional on the stationary cross-section and

transition margins, efficiency also requires that no feasible reassignment would move marginal space toward households who value it more, net of implementation costs. The test below is compensated: reallocate a unit, hold every household at its prior utility with lump-sum transfers, and ask whether goods are left over. A positive residual is a Pareto improvement under any welfare weights.

Proposition 2 (Planner-equilibrium comparison, local). *Consider an interior stationary competitive equilibrium and a planner with the same preferences, housing stock, tenure size menus, stationary young distribution, old-incumbent measure, and transition laws held fixed at the instant of the variation. Suppose policy accounts are resource-neutral and the planner has no social birth payoff, so $\nu_i^g = 0$. Conditional on the equilibrium entry measure and tenure assignments, and away from real menu corners, the competitive allocation solves the planner's conditional allocation problem only if no feasible marginal reallocation of existing floorspace has positive goods-equivalent surplus.*

Let $r_i^F \geq 0$ denote the marginal real resource cost, in consumption-good units, of relaxing young owner i 's financial implementation constraints enough to permit one more unit of owner floorspace. If the planner can directly reassign existing floorspace and relieve financing without using real resources, then $r_i^F = 0$; if it must operate through costly financing instruments, r_i^F prices those real costs; and if the constraint stands for an enforcement primitive no instrument relaxes, the variation is infeasible, the limit $r_i^F = \infty$. For a young owner whose owner menu is slack and whose financial constraints bind, and for an old incumbent at an interior retained-housing choice, the planner's marginal surplus from increasing the young owner's space by one unit and reducing the old incumbent's retained space by one unit is

$$(q + \zeta_i^{O,F} - r_i^F) - (q - \ell_j) = \zeta_i^{O,F} + \ell_j - r_i^F.$$

Thus, in the zero-real-cost finance case, any such pair with $\zeta_i^{O,F} + \ell_j > 0$ prevents the competitive allocation from solving the planner's conditional allocation problem. More generally, a positive young financial access gap is inefficient whenever it can be relaxed at marginal real cost below the value gap relative to the marginal source of space, and a positive retention levy ℓ_j marks an inefficiency whenever the released space has a feasible recipient whose marginal value exceeds the old incumbent's true marginal value.

Conversely, if the financial implementation gaps are zero for all relevant young owners and $\ell_j = 0$ for old incumbents at interior retention choices, then, conditional on entry, tenure assignments, savings, and real menu constraints, the competitive allocation satisfies the planner's conditional housing and fertility first-order conditions for suitable Pareto weights.

Proof. The result is a direct marginal-variation comparison that preserves aggregate floorspace feasibility. Increase household a 's occupied floorspace by dh and reduce household b 's occupied floorspace by the same dh , holding entry, tenure assignments, active real menus, fertility, savings, and stationary transition margins fixed at the instant of the variation. The goods-equivalent first-order welfare effect is

$$(MV_a - \text{marginal real implementation cost for } a) - MV_b.$$

For a young owner with a slack owner menu and binding financial implementation constraints,

$$MV_i^O = q + \zeta_i^{O,F}.$$

If relaxing those constraints has marginal real resource cost r_i^F , the net marginal value of assigning one more unit of space to that owner is $q + \zeta_i^{O,F} - r_i^F$. For an old incumbent at an interior retention choice,

$$MV_j^O = q - \ell_j.$$

Subtracting gives

$$\zeta_i^{O,F} + \ell_j - r_i^F.$$

A positive value is a feasible first-order improvement: transfer goods to hold each household at its pre-variation utility, and the residual $\left[\zeta_i^{O,F} + \ell_j - r_i^F\right] dh$ is available as surplus. Therefore the competitive allocation cannot solve the planner's conditional allocation problem.

If all implementation-relevant financial gaps and retention levies are zero, the competitive marginal values of all households away from real menu corners coincide with the common market user cost q . Real menu corners are respected by both the competitive economy and the constrained planner. With $\nu_i^g = 0$, the competitive fertility condition then matches the planner's fertility condition evaluated at the same scarcity price, and Pareto weights can be chosen to support the competitive consumption allocation. $\square$

The proposition does not characterize the efficiency of the entry, tenure-assignment, or intergenerational transition margins themselves; those margins are held fixed. It also does not say that every cap is a market failure: real menus remain real constraints, while financial gaps matter only to the extent that an implementable instrument can reach them at a low enough real resource cost.

The misallocation has two sides. A young household that cannot post a larger down payment values the marginal unit of space at $q + \zeta_i^{O,F}$, above the market value of that space; the market clears, yet she cannot buy more of it. An old incumbent sitting on a taxable gain keeps the marginal unit while valuing it at $q - \ell_j$, below the market value of that space. Moving one unit from the second household to the first would raise welfare by $\zeta_i^{O,F} + \ell_j - r_i^F$, measured in consumption goods; in the pure-finance case this is $\zeta_i^{O,F} + \ell_j$. Both pieces are computable from data given the preference parameters: the young gap from bundles, via $\zeta_i^{O,F} = \alpha(c_i^O - \chi n_i^O)/(h_i^O - \kappa n_i^O) - q$ when the menu cap is slack, and the old gap from deeds and the tax schedule, via the purchase basis, the current value, the exclusion, and the rate. The old-side force amplifies the misallocation but is not needed for it: the young-access gap is positive whenever a purchase constraint binds with positive shadow value, with or without the gains levy.

Two qualifications matter. First, old occupancy is not inefficient by itself. An old household that keeps a large house because she values it, or because she intends to leave it, is using housing efficiently; those motives are inside u_j^O and the planner counts them. The inefficiency is only the part of retention supported by the forgiveness at death. Second, not every binding constraint is an inefficiency. The size menus are real features of the housing stock: when a renter is pinned at $\bar{h}^R$, the planner—facing the same stock—is pinned with her, and the multiplier on the menu is the shadow value of a real constraint, not a market failure. Changing the menus requires construction, conversion, or reassignment instruments, whose resource costs belong in Ξ^{Π} . Down-payment and payment-to-income limits are financing arrangements; their efficiency cost depends on whether feasible instruments can reach them.

A worked example. A two-type stationary economy puts numbers on every object above. Normalize $q = 1$ and set $\rho = 0.25$, so $P = 4$. The young household has $y = 1$, $a = 0.20$, $B_i = 0$, $\alpha = 0.5$, $\beta = 0.4$, $\kappa = 0.3$, and the equal-scaling child cost $\chi = \kappa q / \alpha = 0.6$; financing is $\phi = 0.80$ and $\psi = 0.30$; the menus are $\bar{h}^R = 0.15$ and $\bar{h}^O = 1$; the discount factor is $\vartheta = 0.2164$, so with $\gamma = 0.5$ and no bequest motive the savings load is $k = \vartheta(1 + \gamma) = 0.325$, and the buyer anticipates no repricing. The old incumbent has $a_j = 0.2$, carried housing $H^0 = 0.25$, $\gamma = 0.5$, no bequest motive, and levy $\ell = 0.2$. The stock $\bar{H} = 5/12$ is chosen so that $q = 1$ is the market-clearing user cost; entry is deterministic with the inside value above the outside option, and the renewal inflow

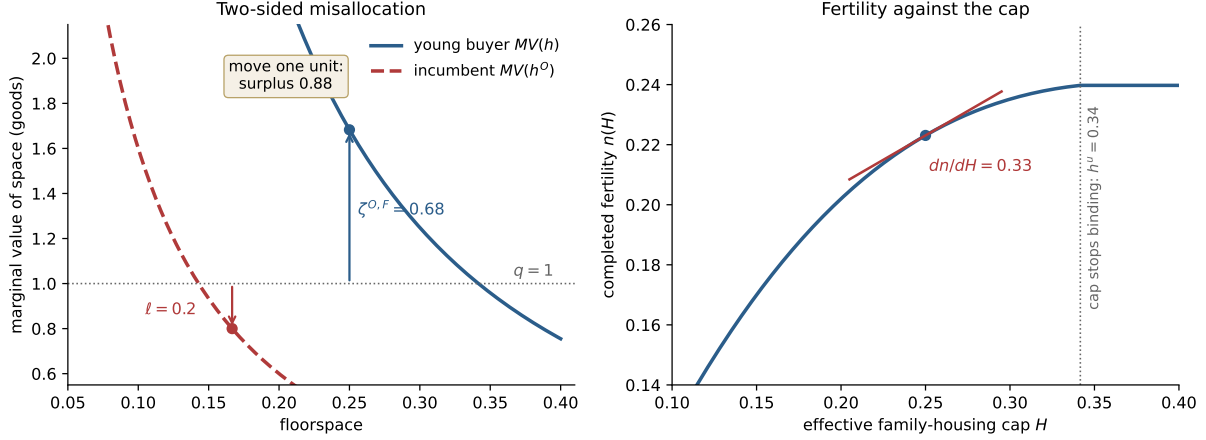


Figure 1: The worked example. Left: marginal value of space. The collateral-capped young owner sits at $h = 0.25$ valuing space at $q + \zeta^{O,F} = 1.68$; the incumbent retains $h^O = 1/6$ valuing it at $q - \ell = 0.80$; moving one unit from her to the buyer frees 0.88 in goods. Right: completed fertility against the effective cap. At the equilibrium cap the slope is $dn/dH = 0.33$, and the cap stops binding at $h^u = 0.34$.

$\bar{R} = 1 - n^*$ sustains the unit cohort.

The young household buys. Both tenures price space at q , so the branch with the larger cap wins: the owner cap is $H^O = \min\{1, a\rho/((1-\phi)q), \psi y/q\} = \min\{1, 0.25, 0.30\} = 0.25$, the collateral component, against the rental menu 0.15. The cap binds—the unconstrained optimum is $h^u = 0.342$ —so $h = 0.25$. The budget is $c + qh + a' = y + a = 1.20$; the savings rule gives $a' = k(c - \chi n) = 0.20$, so the buyer carries exactly her initial wealth, and the fertility condition gives $n^* = 0.223$, with adult consumption $c - \chi n = 0.616$ and adult space $h - \kappa n = 0.183$. The bundle reveals the access gap, $\zeta^{O,F} = \alpha(c - \chi n)/(h - \kappa n) - q = 0.68$: the household would pay sixty-eight percent over market rent for one more unit of space, and cannot post the down payment to get it. The incumbent's released units net $q - \ell$, so her liquidation wealth is $a_j + (q - \ell)H^O = 0.40$ and she keeps $h^O = 1/6$ at the interior solution of $\gamma c_j^O/h_j^O = q - \ell$, valuing retained space at 0.80; her released $1/12$ plus the passive stock $1/6$ house the young buyer, and the market clears at $\bar{H} = 5/12$. The next old cohort carries the young owner's purchase, 0.25, and her savings return $(1+r)a'$, so the stock recirculates; the incumbent's wealth and levy stand in for her cohort's realized repricing, which the buyer does not anticipate. Figure 1 draws the two sides.

Every object can be read off this one allocation: moving one unit of space from the incumbent to the young household frees $\zeta^{O,F} + \ell = 0.88$ in goods; the cap derivative is $dn/dH = 0.33$; and a down-payment grant passes through at $\partial H^O/\partial A_i = \rho/((1-\phi)q) = 1.25$, so a wealth transfer of 0.01 raises the cap by 0.0125 and completed fertility by 0.0041 through access, plus 0.0012 through the budget now that wealth is spendable, 0.0053 in all. The gap, the levy, and the responses are read off the bundle, the tax schedule, and the financing terms.

9 Fertility and entry

The fertility implication does not require a planner who directly values births. It follows from the private household problem. Children require consumption and housing services, so a constraint on

family-capable housing can change fertility even when fertility enters only parental utility.

Fix a tenure branch and suppress the type index. Let H be the effective upper bound on occupied housing in that branch; the user cost is q . For renters, $H = \bar{h}^R$. For young owners,

$$H = \min \left\{ \bar{h}^O, \frac{A_i(x)\rho}{(1-\phi)q}, \frac{\psi y_i}{q} \right\}.$$

Holding $(w, \chi, q, \alpha, \beta, \kappa, k)$ fixed, where $w \equiv y + A_i(x)$ is lifetime resources, the branch problem adds savings to the bundle choice. Concentrating the savings rule $a' = k(c - \chi n)$ into the objective and the budget reduces it to

$$\max_{c, h, n} \{ (1+k) \log(c - \chi n) + \alpha \log(h - \kappa n) + \beta \log n \}$$

up to a type constant, subject to

$$(1+k)(c - \chi n) + \chi n + qh = w, \quad h \leq H, \quad c - \chi n > 0, \quad h - \kappa n > 0 :$$

the static problem with consumption weight $1+k$ and resources w . Each unit of adult consumption commits $1+k$ units of resources, one spent now and k carried for old age. The unconstrained branch allocation is

$$c^u - \chi n^u = \frac{w}{1 + \alpha + \beta + k}, \quad n^u = \frac{\beta w}{(1 + \alpha + \beta + k)(\chi + \kappa q)},$$

$$h^u = \frac{\alpha w}{(1 + \alpha + \beta + k)q} + \kappa \frac{\beta w}{(1 + \alpha + \beta + k)(\chi + \kappa q)}.$$

Suppose $0 < H < h^u$, so the housing cap binds. Then $h(H) = H$, and the constrained fertility choice $n(H)$ is the unique solution to

$$\frac{\beta}{n} = \frac{(1+k)\chi}{w - qH - \chi n} + \frac{\alpha\kappa}{H - \kappa n}.$$

Implicit differentiation gives

$$\frac{dn}{dH} = \frac{\frac{\alpha\kappa}{(H-\kappa n)^2} - \frac{(1+k)\chi q}{(w-qH-\chi n)^2}}{\frac{\beta}{n^2} + \frac{(1+k)\chi^2}{(w-qH-\chi n)^2} + \frac{\alpha\kappa^2}{(H-\kappa n)^2}}.$$

Therefore a relaxation of the binding housing cap raises fertility at H if and only if

$$\alpha\kappa(w - qH - \chi n)^2 > (1+k)\chi q(H - \kappa n)^2.$$

This condition has a direct Stone–Geary interpretation. At an unconstrained interior branch allocation,

$$\frac{q(h^u - \kappa n^u)}{c^u - \chi n^u} = \alpha,$$

unchanged by the savings margin. The child input bundle has the same housing-to-consumption expenditure ratio as the adults' own bundle when

$$\frac{q\kappa}{\chi} = \alpha, \quad \text{equivalently} \quad \chi = \frac{\kappa q}{\alpha},$$

the equal-scaling benchmark. With savings, the threshold for the fertility response sits above it: fertility is strictly increasing in every binding cap relaxation if and only if

$$\chi \leq (1+k) \frac{\kappa q}{\alpha},$$

because forced spending on housing now crowds out saving alongside consumption, and only the share $1/(1+k)$ of the budget hit lands on the adult-consumption margin that prices children. A binding cap implies $u_h/u_c = \alpha(c - \chi n)/(H - \kappa n) > q$, so the condition above holds strictly under equal scaling whenever the cap binds, with room to spare $k\kappa q/\alpha$. Hence, under equal scaling, every relaxation of a strictly binding family-housing cap raises fertility until the cap ceases to bind. If $\chi > (1+k)\kappa q/\alpha$, fertility need not rise from every cap relaxation; in particular, $dn/dH < 0$ for binding caps sufficiently close to the unconstrained optimum. The reason is that a larger cap raises housing slack, which lowers the crowding cost of children, but it also induces the household to buy more housing at price q , reducing consumption. Fertility rises when the slack effect dominates the consumption-cost effect.

This branch result maps into the model's access margins. A relaxation of the rental size cap raises $H = \bar{h}^R$. If the owner down-payment constraint is the unique binding component of the effective owner cap, then

$$\frac{\partial H}{\partial A_i} = \frac{\rho}{(1-\phi)q} > 0.$$

If the payment-to-income constraint is the unique binding component, then

$$\frac{\partial H}{\partial \psi} = \frac{y_i}{q} > 0.$$

The derivative above signs the fertility response to these primitive cap relaxations locally, holding tenure, prices, income, and child costs fixed.

The competitive-equilibrium/planner comparison is a different statement because prices, consumption, tenure assignments, and entry can all change. In the competitive equilibrium, an entered young household in branch m satisfies

$$\frac{\beta (c_i^{CE,m} - \chi n_i^{CE,m})}{n_i^{CE,m}} = \chi + \kappa(q^{CE} + \zeta_i^{CE,m}).$$

A fertility-neutral planner has no direct social birth payoff. If the planner allocation for the same type and branch is interior with respect to any remaining access constraint, then

$$\frac{\beta (c_i^{P,m} - \chi n_i^{P,m})}{n_i^{P,m}} = \chi + \kappa Q^P.$$

Hence

$$n_i^{P,m} > n_i^{CE,m} \iff \frac{c_i^{P,m} - \chi n_i^{P,m}}{c_i^{CE,m} - \chi n_i^{CE,m}} > \frac{\chi + \kappa Q^P}{\chi + \kappa(q^{CE} + \zeta_i^{CE,m})}.$$

Thus a fertility-neutral planner does not mechanically deliver higher fertility merely because the competitive allocation has a positive access value. The branch-level result gives a primitive sufficient condition for a signed fertility response when a policy or planner locally raises a type's effective housing cap holding the branch primitives fixed.

If the planner directly values births through $\mathcal{G}_i(n_i)$, the interior planner condition becomes

$$\frac{\beta(c_i^P - \chi n_i^P)}{n_i^P} + \nu_i^g = \chi + \kappa Q^P.$$

A positive ν_i^g is a separate fertility motive. Holding consumption, the scarcity price of space, and any remaining access constraints fixed, it raises desired fertility. In the full equilibrium, aggregate fertility also depends on prices, consumption, tenure assignments, and entry.

Under the baseline normalization $\bar{n}_i^E = 0$, births generated by the potential-young cohort are

$$N = \bar{M} \int \pi_i^E n_i^Y dG_Y(i).$$

If the outside option has fertility $\bar{n}_i^E$, then cohort births are instead

$$N = \bar{M} \int [\pi_i^E n_i^Y + (1 - \pi_i^E) \bar{n}_i^E] dG_Y(i),$$

and every entry-margin term below replaces n_i^Y by $n_i^Y - \bar{n}_i^E$. In the stationary demographic equilibrium, this birth mass and the renewal inflow satisfy $N + \bar{R} = \bar{M}$; the local comparisons below hold $(\bar{M}, G_Y, F_O, \bar{R})$ fixed and do not by themselves trace the new stationary demographic fixed point after a policy change. Across two allocations,

$$N^P - N^{CE} = \bar{M} \int [n_i^P(\pi_i^P - \pi_i^E) + \pi_i^E(n_i^P - n_i^{CE})] dG_Y(i),$$

where n_i^{CE} is fertility in the competitive-equilibrium inside branch and outside-option fertility is normalized to zero. The second term is the intensive fertility margin among the households who would enter in the competitive equilibrium. The first term is the entry margin induced by the change in inside value. If entry is held fixed, the aggregate fertility effect is just

$$N^P - N^{CE} = \bar{M} \int \pi_i^E (n_i^P - n_i^{CE}) dG_Y(i).$$

With entry held fixed, aggregate fertility rises only where the planner raises conditional fertility for constrained young types. With entry endogenous, higher inside values can also add entrants.

9.1 A local policy decomposition

Let z be a policy parameter. The local fixed-price effect depends on constrained exposure, pass-through to the binding cap, and the fertility response. For renters, $H_i^R = \bar{h}^R$. For owners,

$$H_i^O = \min \left\{ \bar{h}^O, \frac{A_i(x)\rho}{(1-\phi)q}, \frac{\psi y_i}{q} \right\}.$$

Let $m \in \{R, O\}$ denote the active tenure branch and let $\mathcal{E}_m(z)$ be the set of types for whom branch m is uniquely active and $H_i^m(z)$ is the unique binding family-housing cap. Define the policy pass-through to the effective cap by

$$\mathcal{P}_i^m(z) = \frac{\partial H_i^m}{\partial z}.$$

Because the cap is a minimum, a policy that improves a slack component does nothing: a subsidy to units below the binding size does not relax a binding down-payment constraint, and a down-payment

grant does nothing for a renter pinned by the rental menu. If a policy does not relax the binding component of H_i^m , then $\mathcal{P}_i^m(z) = 0$ for this family-space-cap mechanism. Policies that operate primarily through the price q are a different exercise; they are outside this fixed-price formula, not evaluated by it. A transfer of spendable resources also moves fertility through the budget at a fixed cap; the decomposition isolates the access channel, and an instrument that both relaxes the cap and transfers resources adds the standard income term.

For capped types, the fixed-branch fertility response is

$$\Psi_i^m = \frac{\frac{\alpha\kappa}{(h_i^m - \kappa n_i^m)^2} - \frac{\chi q}{(1+k)(c_i^m - \chi n_i^m)^2}}{\frac{\beta}{(n_i^m)^2} + \frac{\chi^2}{(1+k)(c_i^m - \chi n_i^m)^2} + \frac{\alpha\kappa^2}{(h_i^m - \kappa n_i^m)^2}}.$$

If $\chi \leq (1+k)\kappa q/\alpha$ and the cap binds strictly, then $\Psi_i^m > 0$. The boundary case $\chi = (1+k)\kappa q/\alpha$ is equal scaling at lifetime prices: the child's consumption requirement matches its space requirement when adult consumption is priced at its lifetime cost $1+k$.

Let

$$\Theta_i^m \equiv \frac{\partial W_i^m}{\partial H_i^m} = \lambda_i^m \zeta_i^m$$

be the utility value of relaxing the effective cap. Holding prices, tenure choices, the active sets $\mathcal{A}$ —the collection of branch assignments and binding cap components—and the stationary demographic state fixed, the local aggregate fertility effect is

$$\left. \frac{dN}{dz} \right|_{q,o,\mathcal{A}} = \bar{M} \sum_{m \in \{R,O\}} \int_{\mathcal{E}_m(z)} \left[\pi_i^E \Psi_i^m + (n_i^m - \bar{n}_i^E) \frac{\pi_i^E (1 - \pi_i^E)}{\kappa_E} \Theta_i^m \right] \mathcal{P}_i^m(z) dG_Y(i),$$

where $\bar{n}_i^E = 0$ under the baseline outside-option normalization. The first term is the intensive margin: already-entered constrained households adjust fertility by Ψ_i^m per unit of cap relief, positive under $\chi \leq (1+k)\kappa q/\alpha$. The second is the entry margin: relaxing a binding cap raises the value of entering the market, and entrants bring their fertility with them—net of the births they would have had outside. Both margins are gated by the same exposure set $\mathcal{E}_m(z)$: a policy aimed at unconstrained households moves nothing.

At the lifetime equal-scaling point, one observable object drives both margins.

Corollary 1 (Equal scaling and access gaps). *Suppose $\chi = (1+k)\kappa q/\alpha$, equal scaling at lifetime prices, and the effective cap binds strictly. Then*

$$\Psi_i^m = \frac{\kappa}{\alpha} \frac{\zeta_i^m}{c_i^m - \chi n_i^m} \Omega_i^m, \quad \Omega_i^m = \frac{\frac{\alpha}{h_i^m - \kappa n_i^m} + \frac{q}{c_i^m - \chi n_i^m}}{\frac{\beta}{(n_i^m)^2} + \frac{\chi^2}{(1+k)(c_i^m - \chi n_i^m)^2} + \frac{\alpha\kappa^2}{(h_i^m - \kappa n_i^m)^2}} > 0.$$

Under lifetime equal scaling, the access gap ζ_i^m measures both misallocation and fertility exposure. The entry margin runs on the same gap, since $\Theta_i^m = \zeta_i^m / (c_i^m - \chi n_i^m)$. Policies aimed at the largest normalized gaps $\zeta_i^m / (c_i^m - \chi n_i^m)$ raise births the most, holding pass-through, entry, and the weight Ω_i^m fixed.

Proof. At $\chi = (1+k)\kappa q/\alpha$, the numerator of Ψ_i^m is

$$\frac{\alpha\kappa}{(h_i^m - \kappa n_i^m)^2} - \frac{\kappa q^2}{\alpha (c_i^m - \chi n_i^m)^2} = \frac{\kappa}{\alpha} \left(\frac{\alpha}{h_i^m - \kappa n_i^m} - \frac{q}{c_i^m - \chi n_i^m} \right) \left(\frac{\alpha}{h_i^m - \kappa n_i^m} + \frac{q}{c_i^m - \chi n_i^m} \right) :$$

the $1 + k$ in the child cost cancels the $1 + k$ in the consumption term of the numerator. At a binding cap, $\alpha(c_i^m - \chi n_i^m)/(h_i^m - \kappa n_i^m) = q + \zeta_i^m$, so the first parenthesis equals $\zeta_i^m/(c_i^m - \chi n_i^m)$. Dividing by the positive denominator gives the expression above. $\square$

Thus the fixed-branch effect is exposure to a binding family-space constraint, times pass-through to the effective cap, times the fertility response, plus the induced entry response. The decomposition also holds tenure assignments fixed. With deterministic tenure choice, a policy that changes $W_i^O - W_i^R$ moves the indifference boundary $W_i^O = W_i^R$. If the type distribution has density near that boundary and fertility differs across branches, this tenure-switching margin is first order. The formula above is therefore the fixed-tenure component of the local fertility effect. The next subsection characterizes this margin. A full general-equilibrium policy experiment also changes the price, active sets, old housing choices, fiscal transfers, and possibly the resource cost Ξ^Π .

9.2 Tenure switching at the boundary

In this subsection q^R and q^O are tenure-mode user costs that may include wedges around the common baseline cost; the baseline has $q^R = q^O = q$. The results cover branches that differ only in the per-unit user cost and the effective housing cap; tenure-specific amenities, moving costs, or taste shifters require a separate calculation.

Tenure switching is not a price or general-equilibrium effect. It is already present in the fixed-price model because tenure is chosen by

$$W_i^Y = \max\{W_i^R, W_i^O\}.$$

A policy that raises the owner value relative to the renter value moves households near the boundary $W_i^O = W_i^R$ from renting to owning, and because tenure is a discrete choice, the switchers' bundles—and possibly their fertility—jump. Whether they jump depends on a single feature of the environment: whether the two tenures price a unit of space differently. In the baseline, no-arbitrage makes them price it identically, $q^R = q^O = q$, and we show below that the jump is then exactly zero: the local policy formula of the previous subsection is complete. Tenure-cost wedges—a rental premium for large units as in the smooth menu formulation, landlord operating costs, tenure-specific tax treatment—reopen a gap between the per-unit costs of renting and owning, and then the jump is signed by which tenure is the expensive way to occupy space.

The branch problems include savings. Concentrating the rule $a' = k(c - \chi n)$ as above turns each branch into the static problem with consumption weight $1 + k$, and the proof below carries the weight through. The level-set argument also uses that the concentrated continuation adds the same constant in both modes; under the savings rule this holds when induced old-age choices are interior in both, since the continuation is then $(1 + \gamma + \mathbb{E}b_{j'}) \log((1 + r)a')$ plus a draw constant common to the two modes.

Fix prices and suppress the type index. Let A denote the cheaper branch and B the more expensive branch, with tenure-mode user costs

$$q^A < q^B.$$

Let (c_m, h_m, n_m) denote the branch- m optimum, $m \in \{A, B\}$. Let ζ^m be the goods-equivalent value of relaxing the effective family-housing cap in branch m .

Proposition 3 (Fertility at a tenure switch). *Suppose $q^A < q^B$ and the household is indifferent across the two branches, $W^A = W^B$. Then the cheaper branch's effective cap binds with positive*

value, and the switcher into the expensive branch trades consumption for space:

$$h_B > h_A, \quad h_B - \kappa n_B > h_A - \kappa n_A, \quad c_B - \chi n_B < c_A - \chi n_A.$$

If, in addition,

$$\chi \leq (1+k) \frac{\kappa q^A}{\alpha},$$

then $n_B > n_A$.

Proof. The cheaper branch cap must bind with a positive multiplier. If $\zeta^A = 0$, branch A attains the unconstrained value at price q^A . Unconstrained value is strictly decreasing in the user cost, and branch B has value at most its unconstrained value. Hence

$$W^B \leq W_{\text{unc}}^B(q^B) < W_{\text{unc}}^A(q^A) = W^A,$$

contradicting indifference. Thus $h_A = H^A$ and $\zeta^A > 0$.

Next, $h_B > h_A$. If instead $h_B \leq h_A$, the branch- B allocation would be feasible in branch A . At the lower user cost q^A , branch A could choose the same housing and fertility with strictly higher consumption, contradicting $W^A = W^B$.

It remains to order the bundles and sign fertility. Every branch optimum satisfies

$$\frac{\beta}{n} = \frac{\chi}{c - \chi n} + \frac{\alpha \kappa}{h - \kappa n}.$$

Define

$$X = \frac{\chi}{c - \chi n}, \quad Y = \frac{\alpha \kappa}{h - \kappa n}.$$

X is large when adult consumption is squeezed, Y when adult space is squeezed.⁷ Then $n = \beta/(X + Y)$, and branch utility can be written as $K - \Phi(X, Y)$, where K is a type constant and

$$\Phi(X, Y) = (1+k) \log X + \alpha \log Y + \beta \log(X + Y).$$

Tenure indifference puts both branch optima on one level set, $\Phi(X_A, Y_A) = \Phi(X_B, Y_B)$.

The bundle ordering uses no condition on χ . Since $\Phi_X > 0$ and $\Phi_Y > 0$, the level set is a strictly decreasing graph $Y = \varphi(X)$. Occupied housing pins down the position along it: at any branch optimum,

$$h = \kappa \left(\frac{\alpha}{Y} + \frac{\beta}{X + Y} \right) = \kappa \Phi_Y(X, Y).$$

Let $T = X + Y$. Along a level set of Φ ,

$$\left. \frac{d\Phi_Y}{dX} \right|_{\Phi} = \frac{\Phi_{XY}\Phi_Y - \Phi_{YY}\Phi_X}{\Phi_Y},$$

and

$$\Phi_{XY}\Phi_Y - \Phi_{YY}\Phi_X = \frac{(1+k)\alpha}{XY^2} + \frac{(1+k)\beta}{XT^2} + \frac{\alpha\beta X}{Y^2T^2} > 0,$$

so Φ_Y is strictly increasing in X along the level set. Since $h_B > h_A$, we have $X_B > X_A$, and because the graph is decreasing, $Y_B < Y_A$. Hence $c_B - \chi n_B = \chi/X_B < c_A - \chi n_A$ and $h_B - \kappa n_B = \alpha\kappa/Y_B > h_A - \kappa n_A$.

⁷Capital X and Y are local to this proof and unrelated to the bequest share x and income y_i .

Fertility is ordered by where the two optima sit relative to the ray $Y = \alpha X/(1+k)$. Along a level set of Φ ,

$$\text{sign} \left[\frac{d(X+Y)}{dX} \right] = \text{sign}(\alpha X - (1+k)Y),$$

and the ray crosses each level set once: the level set is a decreasing graph, while along the ray $Y = \alpha X/(1+k)$ the value $\Phi = (1+k+\alpha+\beta) \log X + \text{constant}$ is strictly increasing in X . Hence fertility $n = \beta/(X+Y)$ is single-peaked along an indifference curve, rising as the allocation moves toward the ray from the region $(1+k)Y > \alpha X$; the peak sits where the adults' lifetime-priced bundle has the same proportions as the child's, $(h - \kappa n)/(c - \chi n) = (1+k)\kappa/\chi$. At a branch optimum,

$$\frac{\alpha(c_m - \chi n_m)}{h_m - \kappa n_m} = q^m + \zeta^m, \quad \zeta^m \geq 0, \quad \text{so} \quad \frac{Y_m}{X_m} = \frac{\kappa(q^m + \zeta^m)}{\chi}.$$

Under $\chi \leq (1+k)\kappa q^A/\alpha$, both optima lie strictly in the region $(1+k)Y > \alpha X$: branch A because $\zeta^A > 0$, and branch B because $q^B > q^A$. On that region $X+Y$ falls as X rises toward the ray, so $X_B + Y_B < X_A + Y_A$ and $n_B = \beta/(X_B + Y_B) > \beta/(X_A + Y_A) = n_A$. $\square$

At indifference, the only reason to pay more per unit of space is to get more space, so the marginal switcher into the expensive branch always trades consumption for room. When children are at least as space-intensive as their parents' own lifetime-priced bundle ($\chi \leq (1+k)\kappa q^A/\alpha$, the same condition as above, evaluated at the lower user cost), the extra room carries extra children.

The rent-own cases now read off directly, with tenure-mode user costs q^R and q^O that may include wedges on top of the common floorspace cost q . *Baseline* ($q^O = q^R = q$): the modes differ only through their effective caps, and an indifferent household has the same allocation in both—either both caps are slack, or the caps coincide at the binding allocation. Then $n_O = n_R$, there is no discrete jump at the boundary, and the fixed-tenure formula of the previous subsection is the complete fixed-price answer. *Owning dearer* ($q^O > q^R$): ownership is the expensive family-space mode, and an indifferent switcher into ownership upsizes and has $n_O > n_R$. *Renting dearer* ($q^O < q^R$, for example a large-unit rental premium): the inequality reverses; the marginal switcher into ownership moves to a cheaper, smaller unit, and $n_R > n_O$ under the analogous condition at the lower owner user cost. A cost wedge between the tenures is what makes tenure switching a first-order fertility-jump margin.

The switching formula is fixed-price accounting, not a welfare result or a general-equilibrium comparative static. Let types live in a smooth finite-dimensional type space with continuous density $f(i)$, and suppose there are no mass points at the tenure boundary. Let

$$\Delta_i(z) = W_i^O(z) - W_i^R(z), \quad \mathcal{I}(z) = \{i : \Delta_i(z) = 0\},$$

and assume $\nabla_i \Delta_i(z) \neq 0$ on $\mathcal{I}(z)$, with unique branch optima and kinks in active constraints having zero measure. The switching component of the fixed-price fertility effect is

$$\left. \frac{dN^{\text{switch}}}{dz} \right|_{q, \mathcal{A}} = \bar{M} \int_{\mathcal{I}(z)} \pi_i^E (n_i^O - n_i^R) \frac{\partial_z \Delta_i(z)}{\|\nabla_i \Delta_i(z)\|} f(i) dS(i),$$

where $dS(i)$ is surface measure on the indifference boundary. For cap-shifting policies,

$$\partial_z \Delta_i(z) = \Theta_i^O \mathcal{P}_i^O(z) - \Theta_i^R \mathcal{P}_i^R(z), \quad \Theta_i^m = \lambda_i^m \zeta_i^m.$$

Corollary 2 (Owner-access policies). *Consider an owner-access policy with $\mathcal{P}_i^O(z) \geq 0$ and $\mathcal{P}_i^R(z) = 0$. Suppose the condition in Proposition 3 holds for almost every type on $\mathcal{I}(z)$, evaluated*

at the lower user cost. If $q^O > q^R$, the switching component is nonnegative, so the fixed-tenure owner-access formula is a lower bound for this fixed-price fertility channel. If $q^O = q^R$ (the baseline), the switching component is zero. If $q^O < q^R$, the switching component is nonpositive.

Corollary 3 (Rental-menu policies). *Consider a rental-menu policy with $\mathcal{P}_i^R(z) \geq 0$ and $\mathcal{P}_i^O(z) = 0$. Suppose the condition in Proposition 3 holds for almost every type on $\mathcal{I}(z)$, evaluated at the lower user cost. If $q^O > q^R$, the switching component is nonpositive: relaxing the rental cap raises fertility for constrained renters who remain renters, but it can also pull marginal owners back into renting. If $q^O = q^R$ (the baseline), the switching component is zero. If $q^O < q^R$, the switching component is nonnegative.*

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